

A semi-automatic pipeline for transcribing and segmenting child speech

Polychronia Christodoulidou¹, James Tanner², Jane Stuart-Smith², Michael McAuliffe³, Mridhula Murali⁴, Amy Smith⁴, Lauren Taylor⁴, Joanne Cleland⁴, Anja Kuschmann⁴

¹Department of Theoretical & Applied Linguistics, Aristotle University of Thessaloniki, Greece

²English Language & Linguistics, University of Glasgow, UK

³Department of Linguistics, McGill University, Canada

⁴Department of Psychological Sciences & Health, University of Strathclyde, UK

jane.stuart-smith@glasgow.ac.uk

Abstract

This study evaluates both automated transcription (WhisperX) and forced alignment (MFA) in developing a semi-automated pipeline for obtaining acoustic vowel measures from field recordings from 275 children speaking a non-standard, English dialect, Scottish English. As expected, manual correction of speech transcriptions before forced alignment improves the quality of acoustic vowel measures with respect to manually-annotated data, though speech style and recording environment present some challenges for both tools. Adaptation of the MFA pre-trained *english-us-arpa* acoustic model towards the children's speech also improves the quality of acoustic measures, though greater improvement was not found by increasing training sample size.

Index Terms: child speech, automated speech processing, non-standard English, WhisperX, force alignment (MFA)

1. Introduction

Until recently our understanding of child speech production has rested on results from relatively small speaker samples, not least due to difficulties of collecting, curating and analyzing large child speech corpora [1, 2, 3, 4]. Obtaining reliable acoustic phonetic measures from many child speech recordings also encounters numerous challenges. Not only does child speech show more acoustic and speech production variability than adult speech [1, 5, 6], but preparing recordings for measurement, i.e. transcribing and segmenting child speech, can be time consuming [7], and is compounded by many factors including speech style [2] and spoken language variety [3].

Automated speech processing is commonly used for the acoustic phonetic analysis of large adult speech corpora [8], and is now being extended to child speech [2, 3, 7]. Two kinds of automated processing, automated transcription and forced alignment, can reduce data processing time and increase efficiency of working with large speech corpora, though are not without issues: a fully-automated pipeline usually incurs the removal of a large proportion of likely erroneous tokens [3, 8], and tools may be less suitable for low resource populations [9, 10].

Automated transcription tools offer a way to speed up the initial phase of transcribing child speech. Recent automated language assessment tools developed to provide metrics from spontaneous speech interactions often in noisy settings have used cloud-based systems such as *Amazon Transcribe*, without providing transcripts [11, 12]. Open-AI's *Whisper*, and more recently *WhisperX* [13], have the advantage of running locally, which avoids ethical issues around sharing child speech, and are now also being fine-tuned for child speech [14, 15]. To date automated transcription as a precursor for phonetically analysing child speech has not been evaluated.

Forced alignment, i.e. the automated phone labelling and segmentation of speech sounds [3], requires minimally an orthographic transcript of the speech recording, a pronunciation dictionary specifying the phone labels for each word, and sometimes a pre-trained acoustic model specifying acoustic parameters for the phone labels for the variety. Evaluation of forced alignment has been extensively undertaken for adult speech; to date four studies have systematically evaluated forced alignment for child speech (US English: [2, 7], Australian English: [16], NZ English: [3]). Using metrics which assess alignment quality through the degree of overlap between 'gold standard' manually-annotated boundaries and those inserted by aligners, forced alignment of child speech has shown to vary by child age, speech style, whether training data includes the child's dialect, and sound type. To date the Montreal Forced Aligner (MFA) [17] shows the best performance for US [2] and non-standard NZ English [3]. Post-alignment manual correction of segment boundaries is recommended before acoustic measurement [2, 3, 16], though [7]'s results for sibilants based solely on aligned boundaries are encouraging.

This paper aims to evaluate both automated transcription (*WhisperX* [13]) and forced alignment (MFA [17]) in a semi-automated pipeline used to prepare child speech recordings for acoustic vowel analysis. The context is a reference dataset of robust acoustic measures for clinicians and researchers, drawn from field recordings from a cohort of 275 children, who speak a non-standard English dialect. We consider the impact of two interventions during data processing: 1. adjustment of automated transcripts before alignment (Stage 1, Sec. 3), and 2. adaption of acoustic models towards child speech (Stage 2, Sec. 4).

2. Method

The *Variation in Child Speech (VariCS)* [18] project examines variability in children's speech from a cross-sectional and longitudinal perspective. It aims to establish typical ranges for key acoustic measures of respiration, phonation, resonance and articulation, advancing understanding of developmental speech patterns and their variability. Speech data is being collected from 275 children in Scotland, UK, aged five to twelve years, at four time points ('Sweeps') over 18 months [19]. In each sweep, children complete a set of speech tasks, including story retelling, picture description, sentence repetition, single-word production, sound prolongation and rapid syllable repetition. Data collection was conducted in the field, in 23 primary schools across seven councils in Scotland. Recordings were made using a bespoke iPad app that guided children through the activities, and were saved as 32-bit mono WAV files with a 44.1KHz sampling frequency. To minimise background noise and maintain a constant mouth-to-microphone distance, chil-

dren wore a unidirectional head-mounted Shure SM35 microphone.

The data for this study uses the single-word production task collected during Sweep 1. We aimed to collect productions of 20 familiar, easily-imaged words (*teeth, feet, sheep, cake, fish, bed, bat, cat, hat, cup, sock, coat, boat, soap, boot, car, ring, hill, lip, sing*) chosen to capture Scottish English vowels in a CVC structure, plus certain consonants, and two nonwords to consider nasalisation (*meem, beeb*). We used a picture-naming task, repeated three times to investigate variability across repetitions. We anticipated the task would elicit single words. In fact, the speech is neither highly-controlled (though some children only produced the depicted words without other speech or prompts), nor spontaneous (though some children asked questions and/or responded to fieldworker prompts). This stylistic variation, together with other child production features [5, 6], including pitch and voice quality characteristics affecting resonance, as well as other noises, e.g. coughs, sniffs, laughter, with noises from the field recording environment, and fieldworker prompts occurring in 37% recordings (287/783), adds further challenges for automated processing. The speech style thus falls between ‘controlled’ and ‘spontaneous’ speech [2, 3], but is typical of eliciting child speech for acoustic analysis in a speech research/therapeutic context for which greater efficiency from automated processing is highly desirable [4].

After informal testing on 21 recordings from 7 children from a single council area, we found the optimal transcription tool to be WhisperX [13, 20], with the default model (*small*) and language setting to ‘English’, previously trained on 438k hours of adult English [15]. No improvement was found with the *large* model setting [13], adding the target words using `--initial prompt` did not improve output, and although diarization helped for recordings with audible prompts, it was not useful for non-prompted speech, and so was not adopted.

We aimed for a processing pipeline: WhisperX, then MFA, then likely manual checking. After a trial of adjusting automated transcripts for 22 children, the Stage 1 evaluation (Sec. 3) established that robust acoustic vowel measures needed adjustment of all WhisperX output before alignment. Adjusting the WhisperX output was carried out by four phonetically-trained assistants, took ~150 hours and entailed: removing/correcting incorrect transcriptions including hallucinations; adjusting utterance boundaries to reflect children’s speech (sometimes the words were produced like a continuous list, at others, each word was a separate utterance); and separating fieldworker prompts from child speech into distinct tiers. Detailed annotation of non-speech/‘speech-like’ noises could not be made at this scale.

An informal log of adjusting the WhisperX output for 783 recordings from 273 children for Sweep 1 (~13 hours) showed that WhisperX had transcribed all speech (child/fieldworker) for 73% (568/783), though not always correctly and with some instances of ‘hallucinations’ where text was made up. The log also noted the level of adjustment, and showed that: 70% needed ‘a little’ or ‘no’ adjustment (a few words: 56%, none: 13%), 24% needed ‘some’ (several words), and a small proportion required ‘a lot’ (5%) or completely retranscribing (1%). The adjusted WhisperX `.json` output files were then converted into Praat TextGrids using a bespoke Python script [21]. Earlier trialling had shown that utterance-level timestamps gave better input for MFA than WhisperX word-level timestamps.

Following adjustment of most of the WhisperX dataset (654 recordings from 227 children), the Stage 2 evaluation (Sec. 4) ascertained that an adapted version of the MFA *english_us_arpa* acoustic model would substantially improve measurement ac-

curacy. Our choice of MFA for automatic segmentation was informed by its performance for adult non-standard English speech, as well as recent studies for child speech [2, 3]. We selected *english_us_arpa* for pronunciation dictionary and acoustic model because Scottish English is nominally rhotic, and it gave best results for another non-standard English dialect [3]. Before each alignment we validated the corpus and manually expanded the pronunciation dictionary for the children’s productions by adding phonemic transcriptions for Out of Vocabulary items [3, 6].

The role of the two interventions was evaluated by inspecting our intended acoustic measures structured by known phonetic/phonological patterning [9]. Note that the measures presented here, were either extracted from manual segmentation, or directly extracted from force-aligned boundaries without further manual intervention.

Extraction of acoustic measures and processing, for both Stages, followed these steps. First, a Praat script was used to extract the following features: 1. vowel duration, defined in the manual analysis as the interval from start to end of the vowel’s formant structure with normal voicing (excluding breathy or creaky voicing), and 2. the average values of F1 and F2 frequencies from the central 60% of the vowel. Automatically-inserted boundaries were used to define the vowel for measures extracted from MFA-aligned speech. The data were then cleaned and refined in R [22]: for instance, fieldworker speech was removed, any alternative forms of the target words, e.g. *mean* for *meem*, were excluded if the target word was produced, and the final occurrence of any consecutive repeated words was retained.

3. Stage 1: Adjusting WhisperX

The impact of adjusting WhisperX transcripts on quality of forced alignment was examined using a sample of 64 recordings (total duration: 3695s; mean: 58s; 1408 words/vowel phones) from 22 children (aged 5 to 10, mean age: 6.5 years). All speakers except one produced three recordings, corresponding to three repetitions of the target words. The sample was not balanced in terms of gender (10 boys, 12 girls), age (8 five-year-olds, 5 six-year-olds, 3 seven-year-olds, 2 eight-year-olds, 3 nine-year-olds, and 1 ten-year-old), or local dialect area (4 dialects were included). This diversity aimed to assess the performance of forced alignment across the varied dataset.

WhisperX demonstrated a good level of accuracy in transcribing this child speech, with occasional hallucinations. For every target word except *soap*, WhisperX provided at least one alternative form. But WhisperX was also disrupted by noise interference (e.g. background and child noises, whispered fieldworker prompts) and by a particular hyperarticulation found in the speech of many children, extra-long stop closures.

As expected, MFA could align both raw and adjusted WhisperX transcripts, and was particularly effective for cases with minimal interference (e.g., without environmental or children’s noises/soft prompts or very long stops). However, as for WhisperX – and perhaps unsurprisingly given the lack of annotation of non-speech/speech-like noises – MFA could become derailed by noise interference, also when aligning previously-adjusted WhisperX transcripts. The extra-long stop closures also caused problems for MFA when aligning both raw and adjusted WhisperX output; see example in Figure 1.

The performance of MFA was evaluated against manually-segmented files under two conditions: 1. direct application of MFA to unadjusted WhisperX output (**raw**), and 2. application of MFA to adjusted WhisperX output (**adjusted**). Vowel dura-

tion and formant measures were analysed according to the target phonological vowel categories. Our expectation was that measures extracted from MFA alignments from adjusted WhisperX transcripts would give measures closer to those from manual segmentation, than directly from raw WhisperX output.

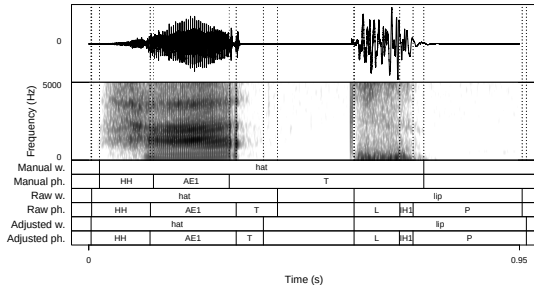


Figure 1: Example of child’s production of hat (followed by lip) with extra-long stop closure (267ms), with manual segmentation (Tiers 1/2), and MFA alignment of words/phones from raw/unadjusted (Tiers 3/4) and adjusted (Tiers 5/6) WhisperX output.

We modelled log-transformed vowel duration ($N = 4018$, excluding tokens without a following consonant) with linear mixed-effect regression (LME) using *lmerTest* [25] in R as a function of method (`{manual, raw, adjusted}`), phonological vowel category, and following consonant voicing (`{voiced, voiceless}`), with vowel $\times$ method and vowel $\times$ voicing interactions, and a random intercept for speaker. We also modelled vowel F1 and F2 with LME, with predictors for method and vowel, and their interaction, with a random intercept for speaker. We used model comparison via likelihood ratio tests (χ^2) to identify significant model terms.

Average vowel durations significantly differ by method ($\chi^2(18) = 55.8, p < 0.0001$). Across vowels, durations from raw WhisperX output (111ms) are significantly shorter from those from manual annotation (129ms; $\hat{\beta} = -0.145, t = -2.64, p < 0.01$); durations from adjusted transcripts do not meaningfully differ from manual segmentation ($\hat{\beta} = -0.08, t = -1.48, p = 0.139$; 118ms). We also observe that durations vary by method within vowel categories ($\chi^2(16) = 45.7, p < 0.0005$). Comparison of estimated durations by method (Figure 2, left) shows that duration differences are vowel-specific in magnitude and direction. Generally speaking, however, the difference between methods is more pronounced for front-high and front-central vowels (/i/, /I/, /e/, /ɛ/).

Figure 2 (right) illustrates the vowel space by method and shows that both raw and adjusted transcripts result in greater variance in F1-F2 space, compared with manual segmentation. Method has a significant effect on the vowel-specific average F1/F2 position (F1: $\chi^2(16) = 67.25, p < 0.0001$; F2: $\chi^2(16) = 197.2, p < 0.0001$). Specifically, /i/ is significantly lowered (higher F1) and retracted (lower F2) for both WhisperX-based methods ($F1_{raw} = 185.24, t = 6.85, p < 0.0001, F1_{adjusted} = 1.47, t = 5.65, p < 0.0001$; $F2_{raw} = -475.13, t = -8.74, p < 0.0001, F2_{adjusted} = -442.15, t = -8.42, p < 0.0001$). Together, the vowel duration and vowel formant results demonstrate that adjusting WhisperX transcripts gives closer phonetic values to manual transcription (compared to no adjustment), though the size of the differences is heavily vowel-specific.

4. Stage 2: Adapting MFA acoustic models

Following adjustment of WhisperX output for 654 recordings from 227 children, we next tried to improve the quality of MFA alignment by shifting the *english_us_arpa* acoustic model towards the children’s variety. A small ‘holdout’ dataset of 9 recordings was manually segmented (age: 5-9 years, mean: 7.2 years; gender: 2 boys/7 girls; total duration: 418s, mean: 46s; 197 words/vowel phones). *mfa_adapt* [26] was used to make three new acoustic models: 1. from the 64 recordings/22 speakers analysed in Stage 1 (adapt22); 2. from 654 recordings/218 speakers, with default `--mapping-tau = 20` (adapt218); 3. from 654 recordings/218 speakers, with `--mapping-tau = 10`, giving greater weight to the input speech (adapt218_tau10).¹ Here, we compare formant measures from manual segmentation against those from MFA alignment from each acoustic model, with the expectation that model adaptation would improve with more speakers.

Given the small number of recordings, we evaluated differences between the acoustic models by comparing the Euclidean distance of formant measures from *english_us_arpa* and each new adapted model, against those from manual segmentation. The greater the Euclidean distance, the further the vowel token derived from the acoustic model is from the manual reference within the F1×F2 vowel space.

To ascertain the influence of model adaptation, we explored the degree of error in formant values between acoustic models by comparing the Euclidean distance, relative to manual segmentation ($N = 788$). Figure 3 (left) illustrates the distribution of Euclidean distances by acoustic model. To get some sense of how accurate each acoustic model was in approximating the manually-annotated formants, we modelled the relative size of the Euclidean distance, where distance values ≤ 50 were selected as sufficiently similar to the manually-annotated values, based on the visual exploration of the distributions and vowel spaces. This measure (≤ 50 vs > 50) was modelled as a logistic regression using *lme4* [27] as a function of acoustic model (`{arpa, adapt22, adapt218, adapt218_tau10}`), and log-scaled duration, and random by-speaker and by-vowel intercepts. We report the estimated marginal difference ($\hat{\Delta}$) between acoustic models using *emmeans* [24]. We find that the probability of a similar distance value is significantly different across acoustic models ($\chi^2(3) = 24.92, p < 0.0001$). Specifically, we find that the *arpa* model performs significantly worse than the other adapted acoustic models ($\hat{\Delta}_{adapt22} = -1.03, z = -3.6, p < 0.002$; $\hat{\Delta}_{adapt218} = -1.2, z = -4.16, p < 0.0002$; $\hat{\Delta}_{adapt218_tau10} = -1.21, z = -4.15, p < 0.0005$), though there are no significant differences between the other acoustic models (Figure 3, right).

5. Discussion

This study reports an evaluation of two key aspects of using automated speech processing to improve the efficiency of preparing 275 children’s field recordings for acoustic analysis [4]. Specifically, we considered the impact of 1. adjusting automated WhisperX transcripts before forced alignment with MFA (Sec. 3), and 2. adapting a pre-trained MFA acoustic model towards the children’s Scottish English dialect (Sec. 4).

WhisperX showed good performance, providing transcripts for 70% of the recordings, and little to no adjustment needed for 73% [20]. As expected, adjustment of WhisperX output be-

¹Adapted acoustic models are available at <https://osf.io/74bsg/>

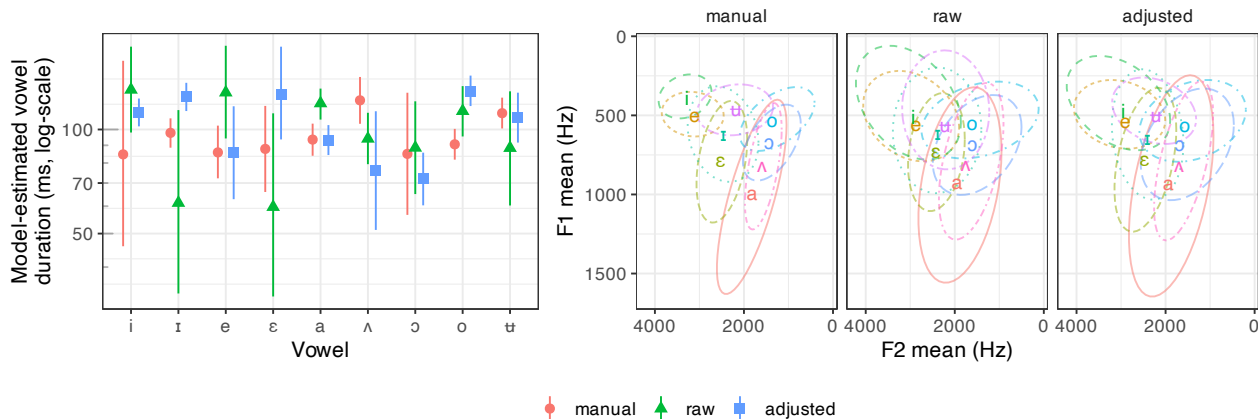


Figure 2: Left: model-estimated vowel durations by vowel phoneme and transcription method. Points and lines indicate estimated means and 95% bootstrapped confidence intervals from 5000 replications [23]. Right: F1/F2 vowel space area by transcription type. Vowel label positions indicate model-estimated marginal mean [24] and ellipses indicate 95% confidence intervals.

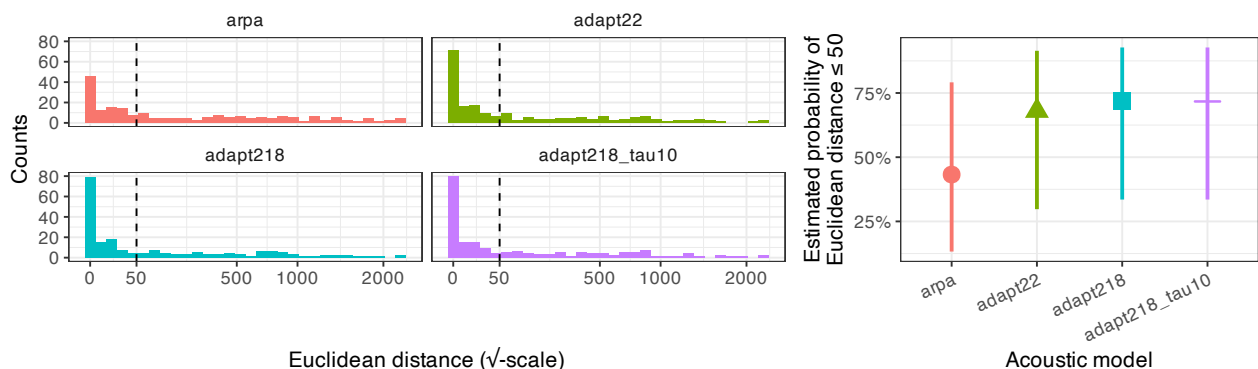


Figure 3: Left: Distribution of Euclidean distances in F1/F2 values between acoustic models. Dashed line shows cutoff for distances ‘sufficiently similar’ to manually-annotated values. Right: Estimated marginal probability of Euclidean distances to be sufficiently close to the manual values by model type. Points and lines indicate estimated marginal mean probability and 95% confidence intervals.

fore alignment improved vowel duration measures with respect to manual annotation. Vowel formant measures obtained from MFA alignment of both raw and adjusted WhisperX transcriptions show the expected distribution in the acoustic vowel space, though with greater variability and more centralized qualities than from manual annotation, especially for /i/.

We also found that both WhisperX and MFA were disrupted by features from the field recording and elicitation task, namely additional noises, and consistent hyperarticulation in this child speech. These findings indicate that this speech style, which is often single words, but may include spontaneous speech too, from field recordings from children speaking a non-standard dialect, does indeed pose challenges for automated processing tools. And this is not surprising: not only are WhisperX’s English *small* model and MFA’s *english_us_arpa* acoustic model trained on adult speech, but the style of training data for both tools is overwhelmingly fluent connected speech, mainly but not entirely spontaneous for Whisper [28], and read speech (*LibriSpeech English*) for MFA. It is unclear whether [20]’s *Flexible* and *Automatic Speech Aligner*, adapted for noisy child speech data, might perform better for our data, given that they evaluate performance via Word Error Rate, rather than the quality of extracted acoustic measures.

At the same time, we were able to improve MFA’s segmentation by adjusting the pretrained acoustic model (*english_us_arpa*) towards the children’s non-standard dialect. The quality of the vowel formant measures was enhanced with respect to manual segmentation [7, 16], and this was achieved with only 22 children in the training data. Of note, the substantial increase to a 218-child training dataset did not give further improvement. This likely is due to all speakers largely saying the same words. More linguistically-varied datasets might need more data to achieve similar performance.

Obtaining robust reference vowel measures from the Sweep 1 dataset required a final phase of manual checking of MFA output [2, 3], which is now completed (a further ~ 150 hours). We note first that the entire manual input is a fraction of estimated time for manual segmentation of speech corpora (up to ~ 800 hours/hour of speech [3, p.183]). Second, the vowel formant measures direct from forced alignment of raw WhisperX output are promising (Figure 2). This suggests that future child speech formant measures could be drawn from an automated approach, as commonly done for adult large-scale speech analysis [8, 2], i.e. by pruning erroneous data against reliable baseline measures obtained by the semi-automatic pipeline developed for this large child speech corpus.

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7. References

- [1] M. E. Beckman, A. R. Plummer, B. Munson, and P. F. Reidy, “Methods for eliciting, annotating, and analyzing databases for child speech development,” *Computer Speech and Language*, vol. 45, pp. 278–299, 9 2017.
- [2] T. J. Mahr, V. Berisha, K. Kawabata, J. Liss, and K. C. Hus-tad, “Performance of forced-alignment algorithms on children’s speech,” *Journal of Speech, Language, and Hearing Research*, vol. 64, pp. 2213–2222, 6 2021.
- [3] R. Fromont, L. Clark, J. W. Black, and M. Blackwood, “Maximiz-ing accuracy of forced alignment for spontaneous child speech,” *Language Development Research*, vol. 3, pp. 182–210, 2023.
- [4] P. Escudero, G. P. Escobar, M. H. Gallego, C. Diskin-Holdaway, and J. Hajek, “A database of multilingual child speech with recordings from a longitudinal project for multilingual education,” in *Nineteenth Australasian International Conference on Speech Science and Technology (SST2024)*, 2024. [Online]. Available: <https://calendly.com/>
- [5] S. Lee, A. Potamianos, and S. Narayanan, “Acoustics of children’s speech: Developmental changes of temporal and spectral param-eters,” *The Journal of the Acoustical Society of America*, vol. 105, pp. 1455–1468, 3 1999.
- [6] M. Benzeghiba, R. D. Mori, O. Deroo, S. Dupont, T. Erbes, D. Jouviet, L. Fissore, P. Laface, A. Mertins, C. Ris, R. Rose, V. Tyagi, and C. Wellekens, “Automatic speech recognition and speech variability: A review,” *Speech Communication*, vol. 49, pp. 763–786, 10 2007.
- [7] T. Knowles, M. Clayards, and M. Sonderegger, “Examining fac-tors influencing the viability of automatic acoustic analysis of child speech,” *Journal of Speech, Language, and Hearing Re-search*, vol. 61, pp. 2487–2501, 2018.
- [8] J. Brand, J. Hay, L. Clark, K. Watson, and M. Sókuthy, “Systematic co-variation of monophthongs across speakers of new zealand english,” *Journal of Phonetics*, vol. 88, p. 101096, 2021. [Online]. Available: <https://doi.org/10.1016/j.wocn.2021.101096>
- [9] S. Babinski, R. Dockum, J. H. Craft, A. Fergus, D. Goldenberg, and C. Bower, “A robin hood approach to forced alignment: English-trained algorithms and their use on australian languages,” *Proceedings of the Linguistic Society of America*, vol. 4, pp. 1–12, 2019.
- [10] E. Chodroff, E. Ahn, and H. Dolatian, “Comparing language-specific and cross-language acoustic models for low-resource phonetic forced alignment,” *Language Doc-umentation and Conservation*, 2024. [Online]. Available: https://www.eleanorchodroff.com/articles/ChodroffAhnDolatian_LangSpecificCrossLangForcedAlignmentComparison.LDC2024.pdf
- [11] C. B. Fox, M. Israelsen-Augenstein, S. Jones, and S. L. Gillam, “An evaluation of expedited transcription methods for school-age children’s narrative language: Automatic speech recognition and real-time transcription,” *Journal of Speech, Language, and Hearing Research*, vol. 64, pp. 3533–3548, 2021. [Online]. Available: <https://doi.org/10.23641/asha>
- [12] L. J. Chaparro-Moreno, H. G. Villasanti, L. M. Justice, J. Sun, and M. B. Schmitt, “Accuracy of automatic processing of speech-language pathologist and child talk during school-based therapy sessions,” *Journal of Speech, Language, and Hearing Research*, vol. 67, pp. 2669–2684, 8 2024.
- [13] M. Bain, J. Huh, T. Han, and A. Zisserman, “Whisperx: Time-accurate speech transcription of long-form audio,” in *Interspeech 2023*, 2023.
- [14] R. Jain, A. Barcovich, M. Yiwere, P. Corcoran, and H. Cucu, *Adaptation of Whisper models to child speech recognition*, 2023. [Online]. Available: <https://arxiv.org/abs/2307.13008v1>
- [15] A. A. Attia, J. Liu, W. Ai, D. Demszky, and C. Espy-Wilson, “Kid-whisper: Towards bridging the performance gap in automatic speech recognition for children vs. adults,” 2024. [Online]. Available: <https://github.com/openai/whisper/discussions/654>
- [16] T. Szalay, M. Shahin, K. Ballard, and B. Ahmed, “Training forced aligners on (mis)matched data: the effect of dialect and age,” in *Proceedings of the Eighteenth Australasian International Conference on Speech Science and Technology (SST 2022)*, 2022, pp. 36–40.
- [17] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, and M. Son-deregger, “Montreal forced aligner: Trainable text-speech align-ment using kaldi,” in *Interspeech*. International Speech and Com-munication Association, 2017, pp. 498–502.
- [18] “Variability in child speech (varics).” [Online]. Available: <https://varics.ac.uk/>
- [19] A. Kuschmann, M. Murali, L. Taylor, A. Smith, J. Cleland, J. Stuart-Smith, and D. Young, *Investigating Variability in Child Speech (VariCS) – the VariCS corpus*, 2024. [Online]. Available: <https://osf.io/wf2r9/>
- [20] D. Liu and J. Xiong, “Fasa: a flexible and automatic speech aligner for extracting high-quality aligned children speech data,” 2024. [Online]. Available: <http://arxiv.org/abs/2406.17926>
- [21] J. Tanner, *Whisper2textgrid.py*. [Online]. Avail-able: <https://github.com/james-tanner/scripts/blob/master/whisper2textgrid.py>
- [22] R Core Team, *R: A Language and Environment for Statistical Computing*, R Foundation for Statistical Computing, Vienna, Austria, 2023. [Online]. Available: <https://www.R-project.org/>
- [23] A. C. Davison and D. V. Hinkley, *Bootstrap Methods and Their Applications*. Cambridge: Cambridge University Press, 1997, iSBN 0-521-57391-2.
- [24] R. V. Lenth, *emmeans: Estimated Marginal Means, aka Least-Squares Means*, 2023, R package version 1.9.0. [Online]. Available: <https://CRAN.R-project.org/package=emmeans>
- [25] A. Kuznetsova, P. B. Brockhoff, and R. H. B. Christensen, “lmerTest package: Tests in linear mixed effects models,” *Journal of Statistical Software*, vol. 82, no. 13, pp. 1–26, 2017.
- [26] M. McAuliffe, *Adapt acoustic model to new data (mfa adapt)*. [Online]. Available: <https://montreal-forced-aligner.readthedocs.io/en/latest/user-guide/workflows/adapt-acoustic-model.html>
- [27] D. Bates, M. Mächler, B. Bolker, and S. Walker, “Fitting linear mixed-effects models using lme4,” *Journal of Statistical Software*, vol. 67, no. 1, pp. 1–48, 2015.
- [28] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. Mcleavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, and J. Scarlett, Eds., vol. 202. PMLR, 23–29 Jul 2023, pp. 28492–28518. [Online]. Available: <https://proceedings.mlr.press/v202/radford23a.html>